

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Nov 2021 P1H Worked Solutions

Nov 2021 | P1H

21 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3****TOPIC: Expanding Brackets**

Nov 2021 P1H - Nov 2021 | P1H

1 (a) Simplify $e^8 \div e^2$
(1)(b) Expand and simplify $(x - 3)(x + 1)$
(2)

(Total for Question 1 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Expanding Brackets**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. (a)

(a) $e^8 \div e^2 = e^6$.

2. (b)

(b) $(x - 3)(x + 1) = x^2 - 2x - 3$.

FINAL ANSWER

(a) e^6 , (b) $x^2 - 2x - 3$

EA

PAST PAPER QUESTION**Q#: 2****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2021 P1H - Nov 2021 | P1H

2 Here is a right-angled triangle.

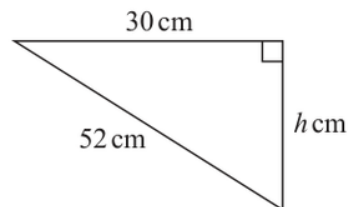


Diagram **NOT**
accurately drawn

Calculate the value of h .

Give your answer correct to 3 significant figures.

$h =$

(Total for Question 2 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 2** **Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**

$$h^2 + 30^2 = 52^2$$

$$h^2 = 52^2 - 30^2 = 1804$$

$$h = \sqrt{1804} = 42.473 \dots$$

FINAL ANSWER

42.5 cm to 3 significant figures.

EA

PAST PAPER QUESTION**Q#: 3****Marks: 4****TOPIC: Probability Toolkit**

Nov 2021 P1H - Nov 2021 | P1H

- 3** There are 54 fish in a tank.
Some of the fish are white and the rest of the fish are red.

Jeevan takes at random a fish from the tank.

The probability that he takes a white fish is $\frac{4}{9}$

- (a) Work out the number of white fish originally in the tank.

.....
(2)

Jeevan puts the fish he took out, back into the tank.
He puts some more white fish into the tank.

Jeevan takes at random a fish from the tank.

The probability that he takes a white fish is now $\frac{1}{2}$

- (b) Work out the number of white fish Jeevan put into the tank.

.....
(2)

.....
(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 4****TOPIC: Probability Toolkit**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate probability**(a) Let the number of white fish be w .

$$\frac{w}{54} = \frac{4}{9}$$

$$w = 54 \times \frac{4}{9} = 24$$

2. (b) There are still 54

(b) There are still 54 fish, and now

$$\frac{\text{white fish}}{54} = \frac{1}{2}$$

$$\text{white fish} = 27$$

3. Jeevan added

Jeevan added

$$27 - 24 = 3$$

FINAL ANSWER

(a) 24 white fish, (b) 3 white fish.

PAST PAPER QUESTION**Q#: 4****Marks: 5****TOPIC: Area & Perimeter**

Nov 2021 P1H - Nov 2021 | P1H

- 4 The diagram shows the front of a wooden door with a semicircular glass window.

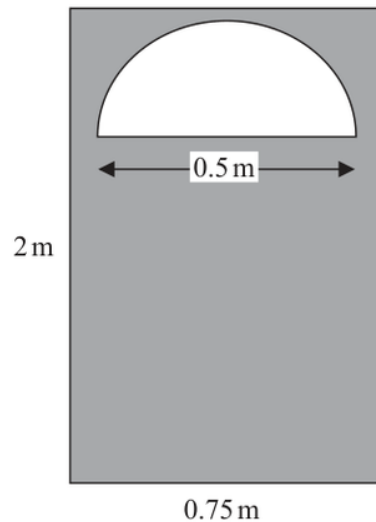


Diagram **NOT**
accurately drawn

Julie wants to apply 2 coats of wood varnish to the front of the door, shown shaded in the diagram.

250 millilitres of wood varnish covers 4 m^2 of the wood.

Work out how many millilitres of wood varnish Julie will need.
Give your answer correct to the nearest millilitre.

..... millilitres

(Total for Question 4 is 5 marks)

PAST PAPER SOLUTION

Q#: 4

Marks: 5

TOPIC: **Area & Perimeter**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Area of the rectangular door

Area of the rectangular door:

$$0.75 \times 2 = 1.5 \text{ m}^2$$

2. The semicircular window has diameter

The semicircular window has diameter 0.5 m, so

$$r = 0.25 \text{ m}$$

3. Area of the semicircular window

Area of the semicircular window:

$$\frac{1}{2}\pi(0.25)^2 = 0.098174\dots \text{ m}^2$$

4. Shaded area for one coat

Shaded area for one coat:

$$1.5 - 0.098174\dots = 1.401825\dots \text{ m}^2$$

5. Calculate area

For 2 coats:

$$2 \times 1.401825\dots = 2.80365\dots \text{ m}^2$$

6. Calculate areaSince 250 ml covers 4 m^2 , varnish needed is

$$\frac{2.80365\dots}{4} \times 250 = 175.22\dots$$

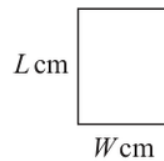
FINAL ANSWER

175 millilitres.

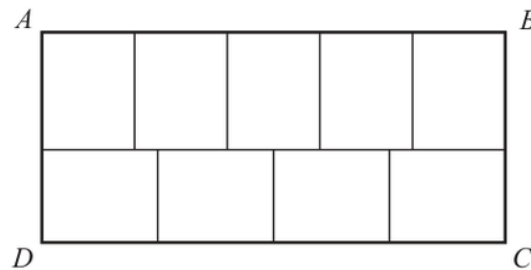
PAST PAPER QUESTION**Q#: 5****Marks: 5****TOPIC: Forming & Solving Equations**

Nov 2021 P1H - Nov 2021 | P1H

- 5 Yasmin has some identical rectangular tiles.
Each tile is L cm by W cm.

Diagram **NOT**
accurately drawn

Using 9 of her tiles, Yasmin makes rectangle $ABCD$, shown in the diagram below.

Diagram **NOT**
accurately drawn

The area of $ABCD$ is 1620 cm^2

Work out the value of L and the value of W .

$L = \dots\dots\dots$ $W = \dots\dots\dots$

(Total for Question 5 is 5 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 5****TOPIC: Forming & Solving Equations**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The top row has 5**

The top row has 5 tile widths and the bottom row has 4 tile lengths, so

$$5W = 4L$$

$$L = \frac{5}{4}W$$

2. Use trigonometry

The rectangle has width $5W$ and height $L + W$:

$$5W(L + W) = 1620$$

3. Substitute

Substitute $L = \frac{5}{4}W$:

$$5W \left(\frac{5}{4}W + W \right) = 1620$$

$$5W \left(\frac{9}{4}W \right) = 1620$$

$$\frac{45}{4}W^2 = 1620$$

$$W^2 = 144$$

$$W = 12, \quad L = \frac{5}{4}(12) = 15$$

FINAL ANSWER

$L = 15$ cm, $W = 12$ cm.

PAST PAPER QUESTION**Q#: 6****Marks: 5****TOPIC: Simultaneous Equations**

Nov 2021 P1H - Nov 2021 | P1H

- 6 Alison buys 5 apples and 3 pears for a total cost of \$1.96
Greg buys 3 apples and 2 pears for a total cost of \$1.22

Michael buys 10 apples and 10 pears.

Work out how much Michael pays for his 10 apples and 10 pears.
Show your working clearly.

\$.....

(Total for Question 6 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 6****Marks: 5****TOPIC: Simultaneous Equations**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve simultaneous equations**Let a be the cost of one apple and p be the cost of one pear.

$$5a + 3p = 1.96$$

$$3a + 2p = 1.22$$

2. Multiply the first equation by

Multiply the first equation by 2:

$$10a + 6p = 3.92$$

3. Multiply the second equation by

Multiply the second equation by 3:

$$9a + 6p = 3.66$$

4. Subtract

Subtract:

$$a = 0.26$$

5. Solve simultaneous equations

Then

$$3(0.26) + 2p = 1.22$$

$$2p = 0.44$$

$$p = 0.22$$

6. Solve simultaneous equations

Michael buys 10 apples and 10 pears:

$$10(0.26) + 10(0.22) = 4.80$$

FINAL ANSWER

\$4.80.

PAST PAPER QUESTION**Q#: 7****Marks: 3****TOPIC: Prime Factors, HCF & LCM**

Nov 2021 P1H - Nov 2021 | P1H

- 7 Write 3.6×10^3 as a product of powers of its prime factors.
Show your working clearly.

(Total for Question 7 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 7** **Marks: 3****TOPIC: Prime Factors, HCF & LCM**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the LCM**

$$3.6 \times 10^3 = 3600$$

$$3600 = 36 \times 100$$

$$36 = 2^2 \times 3^2, \quad 100 = 2^2 \times 5^2$$

2. Find the LCM

So

$$3600 = 2^4 \times 3^2 \times 5^2$$

FINAL ANSWER

$$2^4 \times 3^2 \times 5^2$$

PAST PAPER QUESTION**Q#: 8****Marks: 3****TOPIC: Percentages**

Nov 2021 P1H - Nov 2021 | P1H

- 8** In 2018, the population of Sydney was 5.48 million.
This was 22% of the total population of Australia.

Work out the total population of Australia in 2018
Give your answer correct to 3 significant figures.

..... million

(Total for Question 8 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 8** **Marks: 3**TOPIC: **Percentages**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate percentage**

Sydney's population was 22% of Australia's total population.

$$0.22 \times \text{total} = 5.48$$

$$\text{total} = \frac{5.48}{0.22} = 24.909 \dots$$

2. Calculate percentage

Correct to 3 significant figures,

$$24.909 \dots \approx 24.9$$

FINAL ANSWER

24.9 million

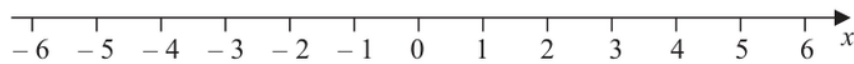
PAST PAPER QUESTION**Q#: 9****Marks: 5****TOPIC: Solving Inequalities**

Nov 2021 P1H - Nov 2021 | P1H

- 9 (i) Solve the inequalities $-7 \leq 2x - 3 < 5$

.....
(3)

- (ii) On the number line, represent the solution set to part (i)



(2)

(Total for Question 9 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 9****Marks: 5****TOPIC: Solving Inequalities**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve inequality**

$$-7 \leq 2x - 3 < 5$$

2. Add 3 to each part

Add 3 to each part:

$$-4 \leq 2x < 8$$

3. Divide by 2

Divide by 2:

$$-2 \leq x < 4$$

4. Use a closed circleOn the number line, use a closed circle at -2 , an open circle at 4 , and shade between them.**FINAL ANSWER**

$$-2 \leq x < 4.$$

PAST PAPER QUESTION**Q#: 10****Marks: 5****TOPIC: Standard & Compound Units**

Nov 2021 P1H - Nov 2021 | P1H

- 10** A solid aluminium cylinder has radius 10 cm and height h cm.

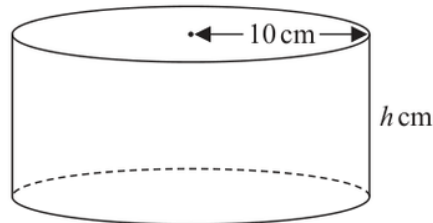


Diagram **NOT**
accurately drawn

The mass of the cylinder is 5.4 kg.
The density of aluminium is 0.0027 kg/cm^3

Calculate the value of h .
Give your answer correct to one decimal place.

$h = \dots\dots\dots$

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 5****TOPIC: Standard & Compound Units**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate volume**

The mass is 5.4 kg, and the density is 0.0027 kg/cm³.

$$\text{volume} = \frac{\text{mass}}{\text{density}} = \frac{5.4}{0.0027} = 2000 \text{ cm}^3$$

2. Evaluate fraction

For the cylinder,

$$V = \pi r^2 h$$

$$2000 = \pi(10)^2 h$$

$$h = \frac{2000}{100\pi} = 6.366 \dots$$

FINAL ANSWER**6.4 cm.**

PAST PAPER QUESTION**Q#: 11****Marks: 5****TOPIC: Cumulative Frequency Diagrams**

Nov 2021 P1H - Nov 2021 | P1H

11 The table gives information about the times taken by 90 runners to complete a 10 km race.

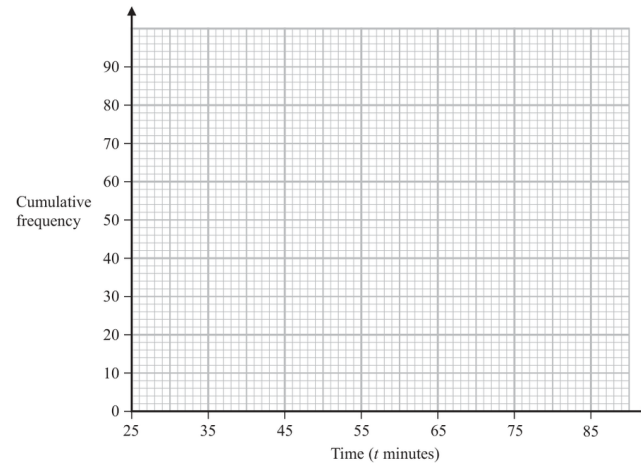
Time (t minutes)	Frequency
$25 < t \leq 35$	12
$35 < t \leq 45$	24
$45 < t \leq 55$	28
$55 < t \leq 65$	12
$65 < t \leq 75$	10
$75 < t \leq 85$	4

(a) Complete the cumulative frequency table.

Time (t minutes)	Cumulative frequency
$25 < t \leq 35$	12
$35 < t \leq 45$	
$45 < t \leq 55$	
$55 < t \leq 65$	
$65 < t \leq 75$	
$75 < t \leq 85$	

(1)

(b) On the grid below, draw a cumulative frequency graph for your table.



(2)

Any runner who completed the race in a time T minutes such that $42 < T \leq 52$ minutes was awarded a silver medal.

(c) Use your graph to find an estimate for the number of runners who were awarded a silver medal.

runners

(2)

(Total for Question 11 is 5 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 5****TOPIC: Cumulative Frequency Diagrams**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The cumulative frequencies are**

The cumulative frequencies are

$$12, 36, 64, 76, 86, 90$$

2. Use cumulative frequencyFor $42 < T \leq 52$,

$$F(52) \approx 36 + \frac{7}{10}(64 - 36) = 55.6$$

$$F(42) \approx 12 + \frac{7}{10}(36 - 12) = 28.8$$

$$55.6 - 28.8 = 26.8$$

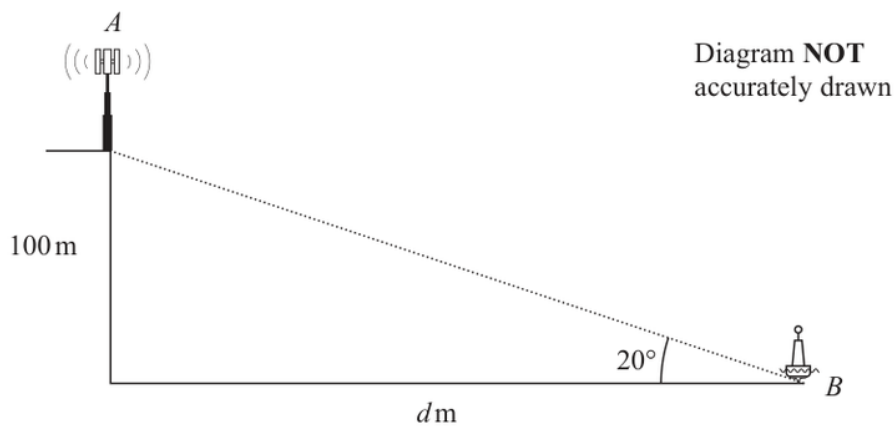
FINAL ANSWER

about 27 runners.

PAST PAPER QUESTION**Q#: 12****Marks: 6****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2021 P1H - Nov 2021 | P1H

- 12** The diagram shows a vertical cliff with a vertical radio mast on top of the cliff and a buoy in the sea.



The height of the cliff is 100 metres.

The buoy is at the point B that is d metres from the base of the cliff.

The angle of elevation from B to the top of the cliff is 20°

- (a) Calculate the value of d .

Give your answer correct to 3 significant figures.

$d = \dots\dots\dots$
(3)

The point A at the top of the radio mast is vertically above the top of the cliff.

The angle of elevation from B to A is 25°

- (b) Calculate the height of the radio mast.

Give your answer correct to 3 significant figures.

$\dots\dots\dots$ m
(3)

(Total for Question 12 is 6 marks)

PAST PAPER SOLUTION**Q#: 12****Marks: 6****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$\tan 20^\circ = \frac{100}{d}$$

$$d = \frac{100}{\tan 20^\circ} = 274.747 \dots$$

2. Use trigonometry

So

$$d = 275 \text{ m}$$

3. (b) The angle of elevation(b) The angle of elevation to the top of the radio mast is 25° .

$$\tan 25^\circ = \frac{100 + h}{274.747 \dots}$$

$$100 + h = 274.747 \dots \tan 25^\circ = 128.116 \dots$$

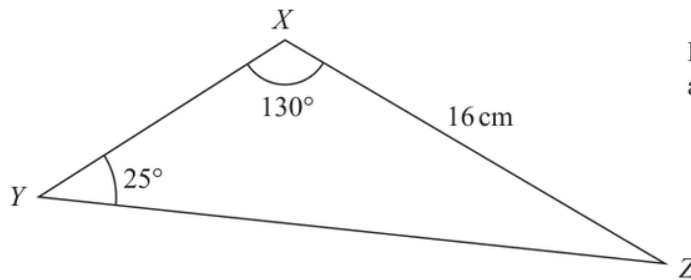
$$h = 28.116 \dots$$

FINAL ANSWER

(a) 275 m, (b) 28.1 m.

PAST PAPER QUESTION**Q#: 13****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Nov 2021 P1H - Nov 2021 | P1H

13 Here is a triangle XYZ .Diagram **NOT**
accurately drawnThe length XZ and the angles YXZ and XYZ are each given correct to 2 significant figures.Calculate the upper bound for the length YZ .

Give your answer correct to one decimal place.

Show your working clearly.

..... cm

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**

Using the sine rule,

$$\frac{YZ}{\sin X} = \frac{XZ}{\sin Y}$$

2. Find upper boundFor the upper bound of YZ :

$$XZ < 16.5, \quad X < 25.5^\circ, \quad Y < 135^\circ$$

$$YZ_{\text{upper}} = \frac{16.5 \sin 25.5^\circ}{\sin 135^\circ} = 10.0457 \dots$$

3. Use trigonometry

Correct to one decimal place,

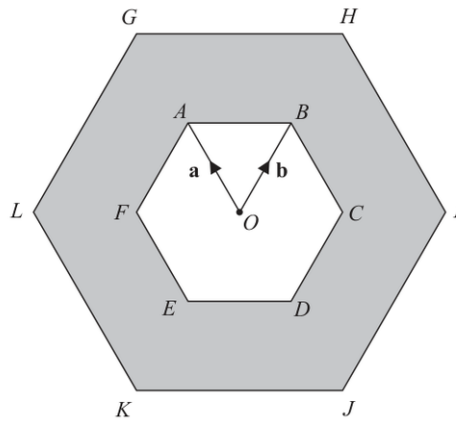
$$10.0457 \dots \approx 10.0$$

FINAL ANSWER

10.0 cm

PAST PAPER QUESTION**Q#: 14****Marks: 8**TOPIC: **Vectors**

Nov 2021 P1H - Nov 2021 | P1H

14 $ABCDEF$ and $GHIJKL$ are regular hexagons each with centre O .Diagram **NOT**
accurately drawn $GHIJKL$ is an enlargement of $ABCDEF$, with centre O and scale factor 2

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

(a) Write the following vectors, in terms of $\mathbf{a}$ and $\mathbf{b}$.
Simplify your answers.

(i) $\vec{AB}$

.....
(1)

(ii) $\vec{KI}$

.....
(2)

(iii) $\vec{LD}$

.....
(2)The triangle OAB has an area of 5 cm^2

(b) Calculate the area of the shaded region.

..... cm^2

(3)

(Total for Question 14 is 8 marks)

PAST PAPER SOLUTION

Q#: 14

Marks: 8

TOPIC: **Vectors**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use vectors

$$\overrightarrow{OA} = \mathbf{a}, \quad \overrightarrow{OB} = \mathbf{b}$$

2. (a)(i)

(a)(i)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \mathbf{b} - \mathbf{a}$$

3. Use vectors

For a regular hexagon,

$$\overrightarrow{OC} = \mathbf{b} - \mathbf{a}, \quad \overrightarrow{OD} = -\mathbf{a}, \quad \overrightarrow{OE} = -\mathbf{b}, \quad \overrightarrow{OF} = \mathbf{a} - \mathbf{b}$$

4. The outer hexagon has scale

The outer hexagon has scale factor 2, so

$$\overrightarrow{OI} = 2(\mathbf{b} - \mathbf{a}), \quad \overrightarrow{OK} = -2\mathbf{b}, \quad \overrightarrow{OL} = 2(\mathbf{a} - \mathbf{b})$$

5. (ii)

(ii)

$$\overrightarrow{KI} = \overrightarrow{OI} - \overrightarrow{OK} = 2(\mathbf{b} - \mathbf{a}) - (-2\mathbf{b}) = 4\mathbf{b} - 2\mathbf{a}$$

6. (iii)

(iii)

$$\overrightarrow{LD} = \overrightarrow{OD} - \overrightarrow{OL} = -\mathbf{a} - 2(\mathbf{a} - \mathbf{b}) = 2\mathbf{b} - 3\mathbf{a}$$

7. (b) The inner hexagon is(b) The inner hexagon is made from 6 equal triangles like OAB :

$$\text{inner area} = 6 \times 5 = 30$$

8. The outer hexagon is anThe outer hexagon is an enlargement with scale factor 2, so area scale factor is $2^2 = 4$:

$$\text{outer area} = 4 \times 30 = 120$$

$$\text{shaded area} = 120 - 30 = 90$$

FINAL ANSWER

(i) $\mathbf{b - a}$, (ii) $\mathbf{4b - 2a}$, (iii) $\mathbf{2b - 3a}$, (b) $\mathbf{90\text{ cm}^2}$.

EA

PAST PAPER QUESTION

Q#: 15

Marks: 8

TOPIC: Probability Diagrams - Venn & Tree Diagrams

Nov 2021 P1H - Nov 2021 | P1H

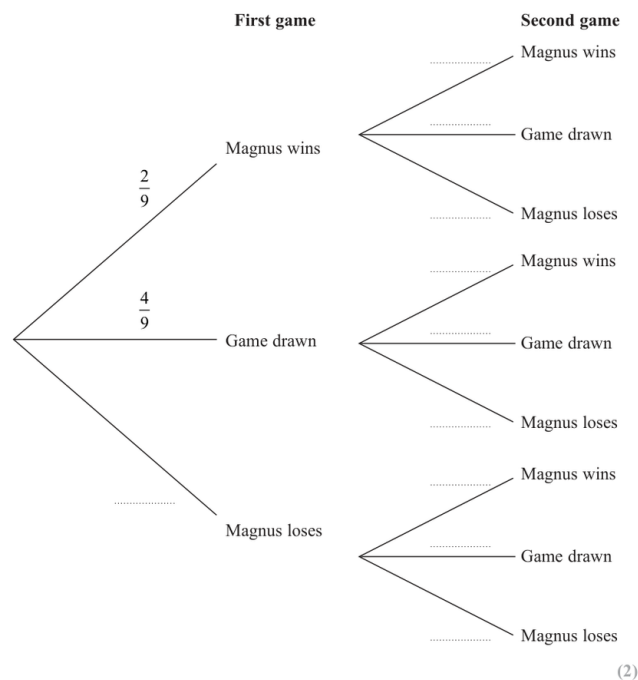
- 15** Magnus and Garry play 2 games of chess against each other.

The probability that Magnus beats Garry in any game is $\frac{2}{9}$

The probability that any game between Magnus and Garry is drawn is $\frac{4}{9}$

The result of any game is independent of the result of any other game.

- (a) Complete the probability tree diagram.



For each game of chess,

the winner gets 2 points and the loser gets 0 points, when the game is drawn, each player gets 1 point.

- (b) Work out the probability that, after 2 games, Magnus and Garry have the same number of points.

(3)

Magnus and Garry now play a third game of chess.

- (c) Work out the probability that, after 3 games, Magnus and Garry have the same number of points.

(3)

(Total for Question 15 is 8 marks)

PAST PAPER SOLUTION**Q#: 15****Marks: 8****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate probability**

For each chess game:

$$P(M \text{ wins}) = \frac{2}{9}, \quad P(\text{draw}) = \frac{4}{9}, \quad P(M \text{ loses}) = \frac{1}{3}$$

2. Calculate probability

After two games, equal points happens if both games are drawn or Magnus wins one and loses one.

$$\begin{aligned} & \left(\frac{4}{9}\right)^2 + 2 \left(\frac{2}{9}\right) \left(\frac{1}{3}\right) \\ &= \frac{16}{81} + \frac{12}{81} = \frac{28}{81} \end{aligned}$$

3. Calculate probability

After three games, equal points means Magnus scores 3 points in total.

4. Calculate probability

This happens with three draws, or with one win, one draw and one loss.

$$\begin{aligned} & \left(\frac{4}{9}\right)^3 + 6 \left(\frac{2}{9}\right) \left(\frac{4}{9}\right) \left(\frac{1}{3}\right) \\ &= \frac{64}{729} + \frac{144}{729} = \frac{208}{729} \end{aligned}$$

FINAL ANSWER

$$\frac{28}{81}, \frac{208}{729}$$

PAST PAPER QUESTION**Q#: 16** **Marks: 5****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Nov 2021 P1H - Nov 2021 | P1H

16 There are 32 students in a class.

In one term these 32 students each took a test in Maths (M), in English (E) and in French (F).

25 students passed the test in Maths.

20 students passed the test in English.

14 students passed the test in French.

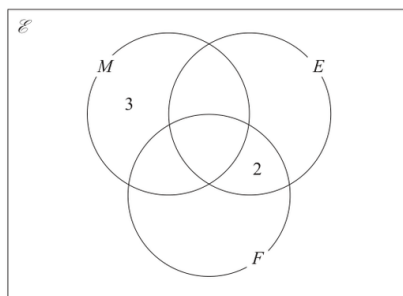
18 students passed the tests in Maths and English.

11 students passed the tests in Maths and French.

4 students failed all three tests.

 x students passed all three tests.

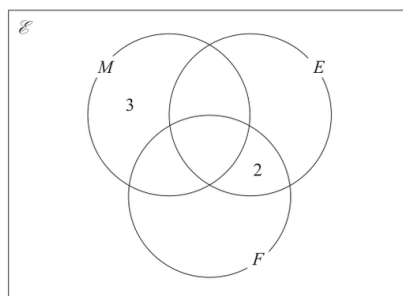
The incomplete Venn diagram gives some more information about the results of the 32 students.



- (a) Use all the given information about the results of students who passed the test in Maths to find the value of x .

 $x = \dots\dots\dots$
 (2)

- (b) Use your value of x to complete the Venn diagram to show the number of students in each subset.



(2)

A student who passed the test in Maths is chosen at random.

- (c) Find the probability that this student failed the test in French.

 $\dots\dots\dots$
 (1)
(Total for Question 16 is 5 marks)

PAST PAPER SOLUTION**Q#: 16****Marks: 5****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use the Maths circle**

Use the Maths circle.

$$M \text{ only} = 3$$

$$M \cap E \text{ only} = 18 - x$$

$$M \cap F \text{ only} = 11 - x$$

2. The total in isThe total in M is 25:

$$3 + (18 - x) + (11 - x) + x = 25$$

$$32 - x = 25$$

$$x = 7$$

3. Calculate probability

So:

$$M \cap E \text{ only} = 11, \quad M \cap F \text{ only} = 4, \quad E \cap F \text{ only} = 2$$

$$E \text{ only} = 20 - 11 - 2 - 7 = 0$$

$$F \text{ only} = 14 - 4 - 2 - 7 = 1$$

4. Calculate probabilityFor a student who passed Maths, failed French means M but not F :

$$\frac{3 + 11}{25} = \frac{14}{25}$$

FINAL ANSWER

$$x = 7, \text{ and } \frac{14}{25}.$$

PAST PAPER QUESTION**Q#: 17****Marks: 8****TOPIC: Rearranging Formulas**

Nov 2021 P1H - Nov 2021 | P1H

17 (a) Factorise $6y^2 - y - 5$
(2)(b) Make f the subject of $w = \frac{2f + 3}{8 - f}$
(3)(c) Express $4x^2 - 8x + 7$ in the form $a(x + b)^2 + c$ where a , b and c are integers......
(3)

(Total for Question 17 is 8 marks)

PAST PAPER SOLUTION**Q#: 17****Marks: 8****TOPIC: Rearranging Formulas**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$6y^2 - y - 5 = (6y + 5)(y - 1)$$

2. (b)

(b)

$$w = \frac{2f + 3}{8 - f}$$

$$w(8 - f) = 2f + 3$$

$$8w - 3 = f(w + 2)$$

$$f = \frac{8w - 3}{w + 2}$$

3. (c)

(c)

$$4x^2 - 8x + 7 = 4(x^2 - 2x) + 7$$

$$= 4((x - 1)^2 - 1) + 7$$

$$= 4(x - 1)^2 + 3$$

FINAL ANSWER

$$(6y + 5)(y - 1), f = \frac{8w - 3}{w + 2}, 4(x - 1)^2 + 3$$

PAST PAPER QUESTION

Q#: 18

Marks: 3

TOPIC: **Fractions, Decimals & Percentages**

Nov 2021 P1H - Nov 2021 | P1H

18 $0.4\dot{x}$ is a recurring decimal.

x is a whole number such that $1 \leq x \leq 9$

Find, in terms of x , the recurring decimal $0.4\dot{x}$ as a fraction.

Give your fraction in its simplest form.

Show clear algebraic working.

(Total for Question 18 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 18****Marks: 3****TOPIC: Fractions, Decimals & Percentages**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let**

Let

$$y = 0.4\dot{x}$$

2. Simplify fraction

So

$$y = 0.4xxx \dots$$

3. Simplify fraction

Multiply by 10 and 100:

$$10y = 4.xxx \dots$$

$$100y = 4x.xxx \dots$$

4. Subtract

Subtract:

$$100y - 10y = (4x.xxx \dots) - (4.xxx \dots)$$

$$90y = x + 36$$

5. Simplify fraction

Therefore

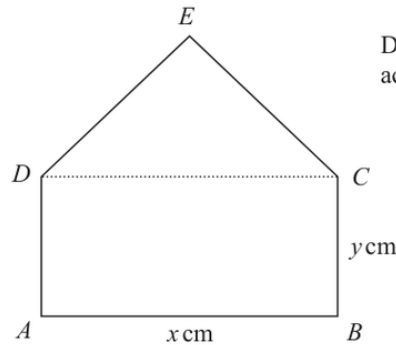
$$y = \frac{x + 36}{90}$$

FINAL ANSWER

$$\frac{x+36}{90}$$

PAST PAPER QUESTION**Q#: 19****Marks: 6****TOPIC: Algebraic Proof**

Nov 2021 P1H - Nov 2021 | P1H

19 $ABCED$ is a five-sided shape.Diagram **NOT**
accurately drawn

$ABCD$ is a rectangle.
 CED is an equilateral triangle.

$AB = x \text{ cm}$ $BC = y \text{ cm}$

The perimeter of $ABCED$ is 100 cm.

The area of $ABCED$ is $R \text{ cm}^2$

(a) Show that $R = \frac{x}{4} \left(200 - [6 - \sqrt{3}]x \right)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$x = \dots\dots\dots$
 (2)

(ii) Explain why the maximum value of R is given by this value of x .

(1)

(Total for Question 19 is 6 marks)

PAST PAPER SOLUTION**Q#: 19****Marks: 6****TOPIC: Algebraic Proof**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Rearrange formula**

The perimeter is made from three sides of length x and two sides of length y :

$$3x + 2y = 100$$

2. Rearrange formula

So

$$y = \frac{100 - 3x}{2}$$

3. The area is the area

The area is the area of the rectangle plus the area of the equilateral triangle:

$$R = xy + \frac{\sqrt{3}}{4}x^2$$

4. Substitute

Substitute $y = \frac{100 - 3x}{2}$:

$$R = x \left(\frac{100 - 3x}{2} \right) + \frac{\sqrt{3}}{4}x^2$$

$$R = 50x - \frac{3}{2}x^2 + \frac{\sqrt{3}}{4}x^2$$

$$R = \frac{x}{4} (200 - (6 - \sqrt{3})x)$$

5. Rearrange formula

For the maximum, write

$$R = -\frac{6 - \sqrt{3}}{4}x^2 + 50x$$

6. The maximum occurs at

The maximum occurs at

$$x = \frac{-b}{2a} = \frac{-50}{2 \left(-\frac{6 - \sqrt{3}}{4} \right)}$$

$$x = \frac{100}{6 - \sqrt{3}}$$

7. Solve quadratic equation

The coefficient of x^2 is negative, so the quadratic graph opens downwards and this vertex gives a maximum.

FINAL ANSWER

$$R = \frac{x}{4} (200 - (6 - \sqrt{3})x), \text{ and } x = \frac{100}{6 - \sqrt{3}}.$$

EA

PAST PAPER QUESTION**Q#: 20****Marks: 7****TOPIC: Coordinate Geometry**

Nov 2021 P1H - Nov 2021 | P1H

- 20** The straight line **L** passes through point $A(-6, 2)$ and point $B(5, 3)$
The straight line **M** is perpendicular to **L** and passes through the midpoint of A and B .
The line **M** intersects the line $x = -1$ at point C .

Calculate the area of triangle ABC .

(Total for Question 20 is 7 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 7****TOPIC: Coordinate Geometry**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$A = (-6, 2), \quad B = (5, 3)$$

2. The midpoint of isThe midpoint of AB is

$$\left(\frac{-6 + 5}{2}, \frac{2 + 3}{2} \right) = \left(-\frac{1}{2}, \frac{5}{2} \right)$$

3. The gradient of isThe gradient of AB is

$$\frac{3 - 2}{5 - (-6)} = \frac{1}{11}$$

4. Find the gradient

So the perpendicular gradient is

$$-11$$

5. Evaluate fractionLine M passes through $\left(-\frac{1}{2}, \frac{5}{2}\right)$, so

$$y - \frac{5}{2} = -11 \left(x + \frac{1}{2} \right)$$

6. Evaluate fractionAt point C , $x = -1$:

$$y - \frac{5}{2} = -11 \left(-\frac{1}{2} \right)$$

$$y - \frac{5}{2} = \frac{11}{2}$$

$$y = 8$$

7. Calculate value

So

$$C = (-1, 8)$$

8. Use vectors

Now use vectors from A :

$$\vec{AB} = (11, 1), \quad \vec{AC} = (5, 6)$$

9. Area of triangle is

Area of triangle ABC is

$$\frac{1}{2}|11(6) - 1(5)|$$

$$= \frac{1}{2}(61)$$

$$= 30.5$$

FINAL ANSWER

30.5

EA

PAST PAPER QUESTION**Q#: 20****Marks: 7****TOPIC: Coordinate Geometry**

Nov 2021 P1H - Nov 2021 | P1H

- 20** The straight line **L** passes through point $A(-6, 2)$ and point $B(5, 3)$
The straight line **M** is perpendicular to **L** and passes through the midpoint of A and B .
The line **M** intersects the line $x = -1$ at point C .

Calculate the area of triangle ABC .

(Total for Question 20 is 7 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 7****TOPIC: Coordinate Geometry**

Nov 2021 P1H - Nov 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$A = (-6, 2), \quad B = (5, 3)$$

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$$y - \frac{5}{2} = \frac{11}{2}$$

$$y = 8$$

7. Calculate value

So

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Now use vectors from A :

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$$= \frac{1}{2}(61)$$

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FINAL ANSWER

30.5

EA